

Транспонирование матриц:

1) □

$$\lambda \tau = (A^T)^T = \left(\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}^T \right)^T =$$

$$= \begin{pmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \dots & \dots & \dots & \dots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix}^T = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} = A = \text{нр.}\tau \quad \square$$

2) □

$$\lambda \tau = (\lambda \cdot A)^T = \left(\lambda \cdot \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \right)^T =$$

$$= \begin{pmatrix} \lambda \cdot a_{11} & \lambda \cdot a_{12} & \dots & \lambda \cdot a_{1n} \\ \lambda \cdot a_{21} & \lambda \cdot a_{22} & \dots & \lambda \cdot a_{2n} \\ \dots & \dots & \dots & \dots \\ \lambda \cdot a_{m1} & \lambda \cdot a_{m2} & \dots & \lambda \cdot a_{mn} \end{pmatrix}^T =$$

$$= \begin{pmatrix} \lambda \cdot a_{11} & \lambda \cdot a_{21} & \dots & \lambda \cdot a_{m1} \\ \lambda \cdot a_{12} & \lambda \cdot a_{22} & \dots & \lambda \cdot a_{m2} \\ \dots & \dots & \dots & \dots \\ \lambda \cdot a_{1n} & \lambda \cdot a_{2n} & \dots & \lambda \cdot a_{mn} \end{pmatrix} = \left(\lambda \cdot \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix} \right)^T =$$

$$= \lambda \cdot A^T = \text{нр.}\tau \quad \square$$

$$3) \square$$

$$A+B)^T = \left(\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{pmatrix} \right)^T =$$

$$= \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \dots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & \dots & a_{2n}+b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & \dots & a_{mn}+b_{mn} \end{pmatrix}^T =$$

$$= \begin{pmatrix} a_{11}+b_{11} & a_{21}+b_{21} & \dots & a_{m1}+b_{m1} \\ a_{12}+b_{12} & a_{22}+b_{22} & \dots & a_{m2}+b_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n}+b_{1n} & a_{2n}+b_{2n} & \dots & a_{mn}+b_{mn} \end{pmatrix} =$$

$$= \begin{pmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{21} & \dots & b_{m1} \\ b_{12} & b_{22} & \dots & b_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1n} & b_{2n} & \dots & b_{mn} \end{pmatrix} =$$

$$= A^T + B^T = (A+B)^T$$

$\square$

4) □

$$np.z = (A \cdot B)^T = \left(\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{pmatrix} \right)^T =$$

$$= \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + \dots + a_{1n}b_{n1} & a_{11}b_{12} + a_{12}b_{22} + \dots + a_{1n}b_{n2} & \dots & a_{11}b_{1n} + a_{12}b_{2n} + \dots + a_{1n}b_{nn} \\ a_{21}b_{11} + a_{22}b_{21} + \dots + a_{2n}b_{n1} & a_{21}b_{12} + a_{22}b_{22} + \dots + a_{2n}b_{n2} & \dots & a_{21}b_{1n} + a_{22}b_{2n} + \dots + a_{2n}b_{nn} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}b_{11} + a_{m2}b_{21} + \dots + a_{mn}b_{n1} & a_{m1}b_{12} + a_{m2}b_{22} + \dots + a_{mn}b_{n2} & \dots & a_{m1}b_{1n} + a_{m2}b_{2n} + \dots + a_{mn}b_{nn} \end{pmatrix}^T =$$

$$= \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + \dots + a_{1n}b_{n1} & a_{21}b_{11} + a_{22}b_{21} + \dots + a_{2n}b_{n1} & \dots & a_{m1}b_{11} + a_{m2}b_{21} + \dots + a_{mn}b_{n1} \\ a_{11}b_{12} + a_{12}b_{22} + \dots + a_{1n}b_{n2} & a_{21}b_{12} + a_{22}b_{22} + \dots + a_{2n}b_{n2} & \dots & a_{m1}b_{12} + a_{m2}b_{22} + \dots + a_{mn}b_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{11}b_{1n} + a_{12}b_{2n} + \dots + a_{1n}b_{nn} & a_{21}b_{1n} + a_{22}b_{2n} + \dots + a_{2n}b_{nn} & \dots & a_{m1}b_{1n} + a_{m2}b_{2n} + \dots + a_{mn}b_{nn} \end{pmatrix} =$$

$$= \begin{pmatrix} b_{11}a_{11} + b_{21}a_{12} + \dots + b_{n1}a_{1n} & b_{11}a_{21} + b_{21}a_{22} + \dots + b_{n1}a_{2n} & \dots & b_{11}a_{m1} + b_{21}a_{m2} + \dots + b_{n1}a_{mn} \\ b_{12}a_{11} + b_{22}a_{12} + \dots + b_{n2}a_{1n} & b_{12}a_{21} + b_{22}a_{22} + \dots + b_{n2}a_{2n} & \dots & b_{12}a_{m1} + b_{22}a_{m2} + \dots + b_{n2}a_{mn} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1n}a_{11} + b_{2n}a_{12} + \dots + b_{nn}a_{1n} & b_{1n}a_{21} + b_{2n}a_{22} + \dots + b_{nn}a_{2n} & \dots & b_{1n}a_{m1} + b_{2n}a_{m2} + \dots + b_{nn}a_{mn} \end{pmatrix} =$$

$$= \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{pmatrix} \cdot \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} = B^T \cdot A^T = np.z$$

□